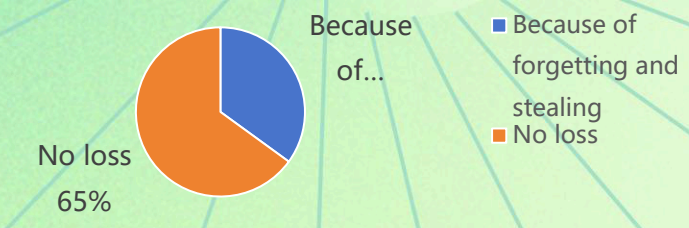
Smart Umbrella

Core issues： Easy to lose: 35% of umbrellas are lost each year due to forgetting or theft.



Security hazard: 50% of users require handheld lighting on rainy days, which is inconvenient to operate.





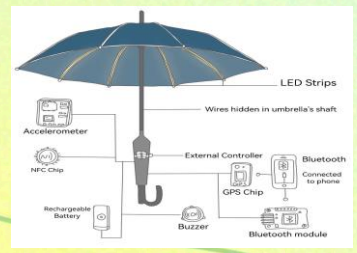
Single functionality:Most of the umbrellas on the market now lack interactivity and cannot be intelligent.



Multifunctional umbrellas on the market



Solution： Combining loss prevention, safety, and entertainment to create a new experience for rainy weather travel.



Our design idea is a diversified Modular Intelligent umbrella

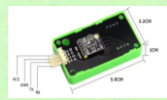
Functional modules	technical realization	user value
Intelligent Loss Prevention	GPS real-time positioning+NFC identity binding+buzzer alarm	Reduce loss rate by 60%
adaptive lighting	LED strip+fiber optic light guide+light sensor	Improved visibility in low light environments by 85%
Abnormal behavior detection	Accelerometer monitoring for drops/abnormal shaking	Instantly trigger an alarm to prevent theft



Acceleration button module



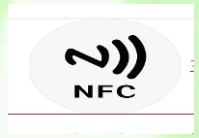
GPS module





LED and lighting module





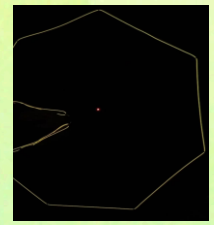
NFC read and input module

contribution

Jiayi Peng	25%	Hardware Selection and Procurement, Circuit Design
Haiqi Shen	25%	Embedded Software Development and Function Integration
Mengxuan Shen	25%	User Testing, Data Analysis, and Report Writing
Chenxin Li	25%	Industrial Design, Waterproof Structure and Appearance Optimization

test result

Satisfaction with loss prevention function: 92% of users think the alarm is timely and effective
Lighting uniformity: 88% of users approve the safety of low light environment.







feedback

*User Wang * *: "I am no longer afraid to lose my umbrella in rainy days. LED lighting can also be used as a flashlight! ”
*User Li * *: " With the anti loss function, I am not afraid of losing my umbrella! ”

Future upgrade direction

Lightweight design: optimized umbrella handle structure, reducing weight by 20%
Energy Innovation: Integrated Solar Thin Film Charging.

The multifunctional umbrella in the future must be more imaginative



Intelligent Expansion:Gesture opening and closing control (triggered by swinging)
Linkage with smart homes (automatic window closing on rainy days).